Programming Fundamentals

LAB-03-04 Submission

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Roll No: **09**

Program: **Computer Science**

Section: **(E)**

Submitted to: **Sir Asad Majeed**

Date: **18/10/2026**

TASK. 1.

Assume the following variable are defined

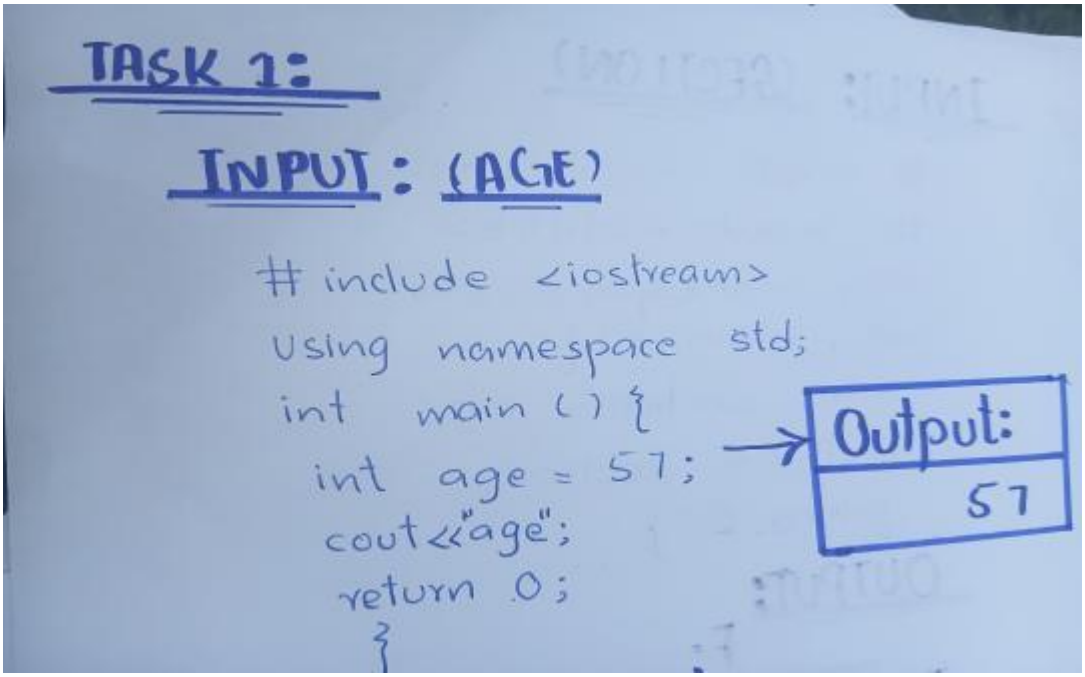
int. age

double. pay.

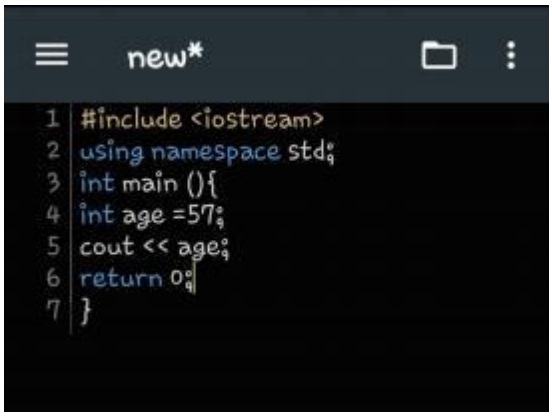
char. section

write a single can statement that will read input into each of these variable.

HAND WRITTEN CODE **(AGE)**



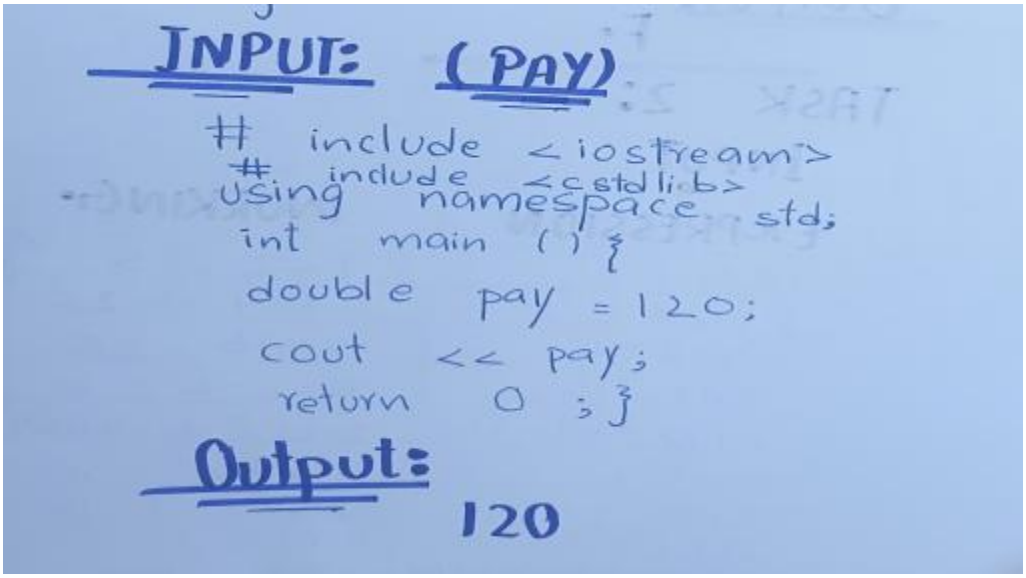
CODE. WRITTEN ON PHONE



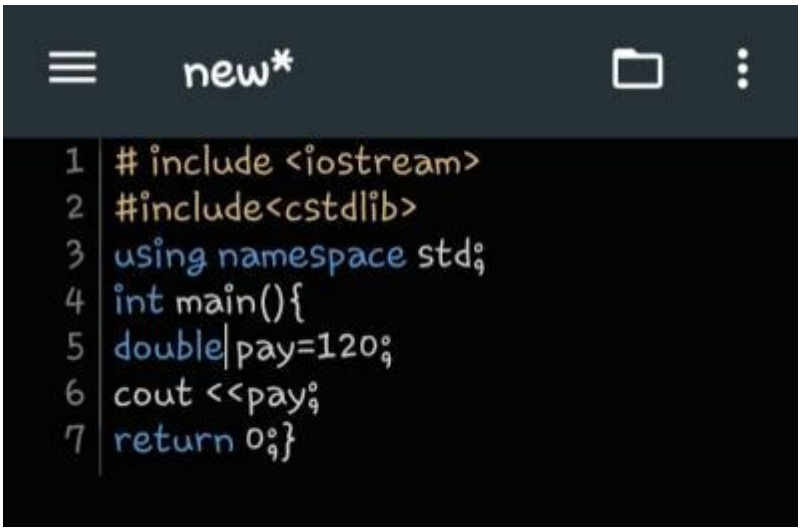
OUTPUT



HAND WRITTEN CODE. **(PAY)**



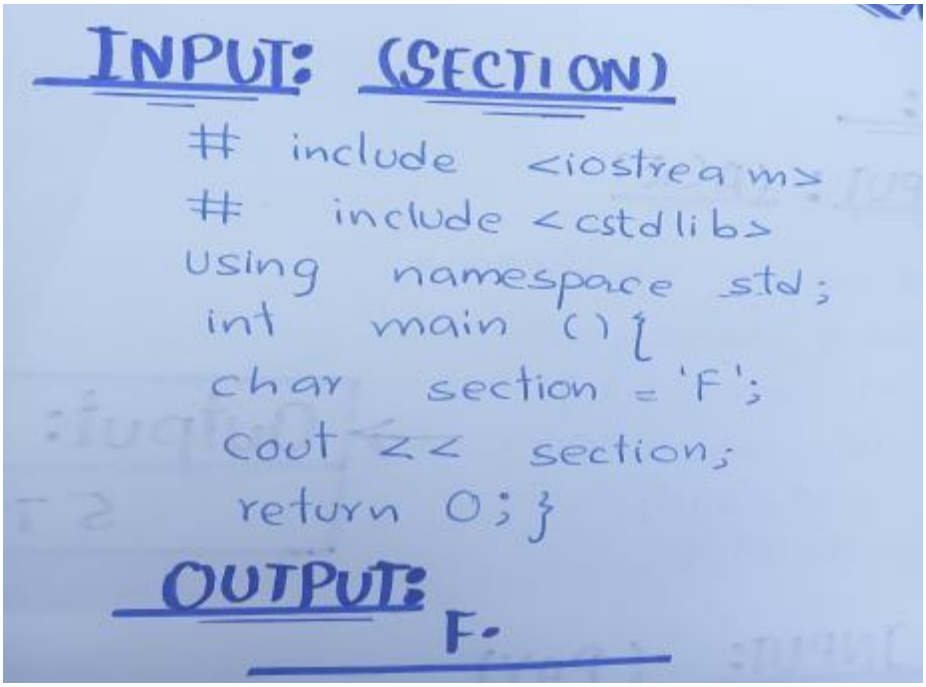
CODE. WRITTEN ON PHONE



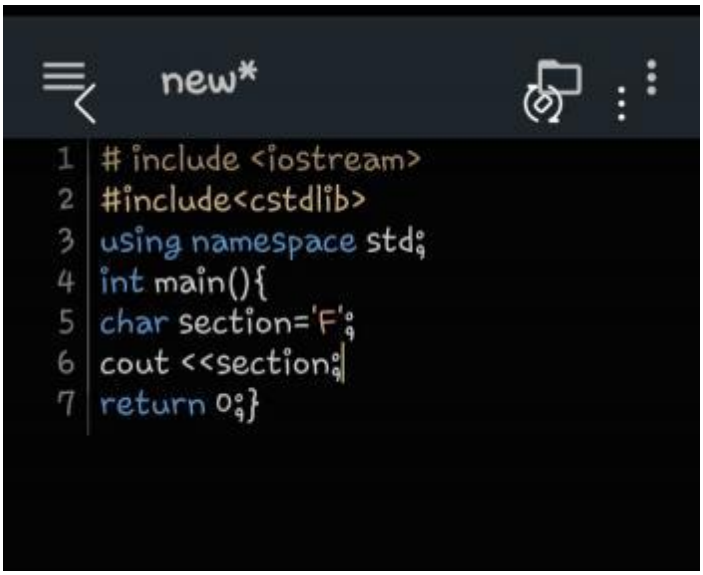
OUTPUT



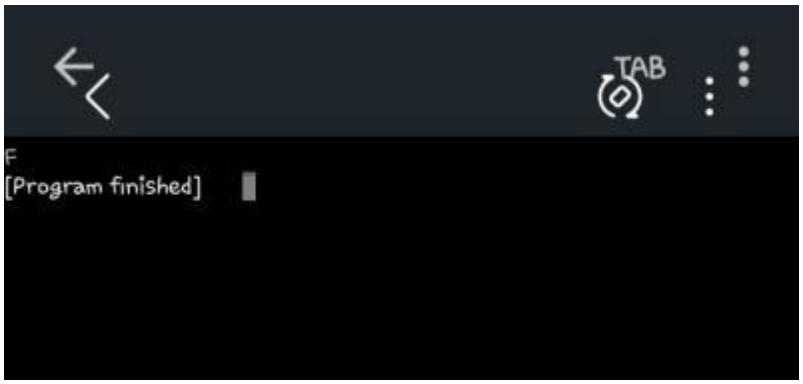
HAND WRITTEN CODING. **(SECTION)**



CODE WRITTEN ON PHONE



OUTPUT

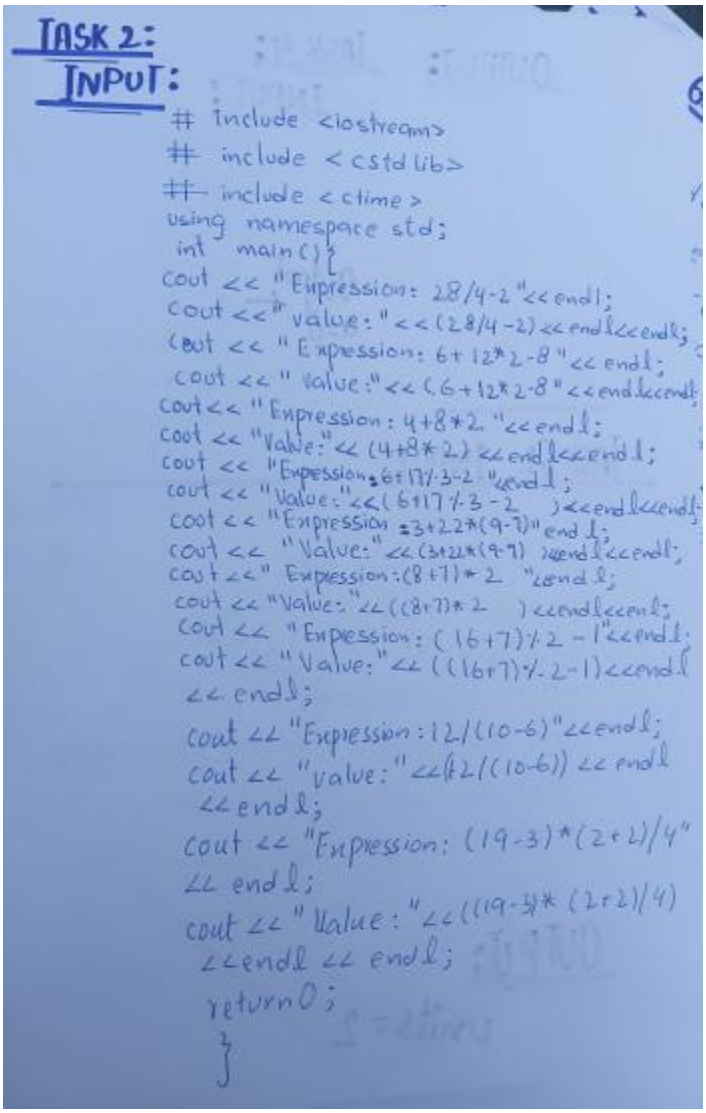


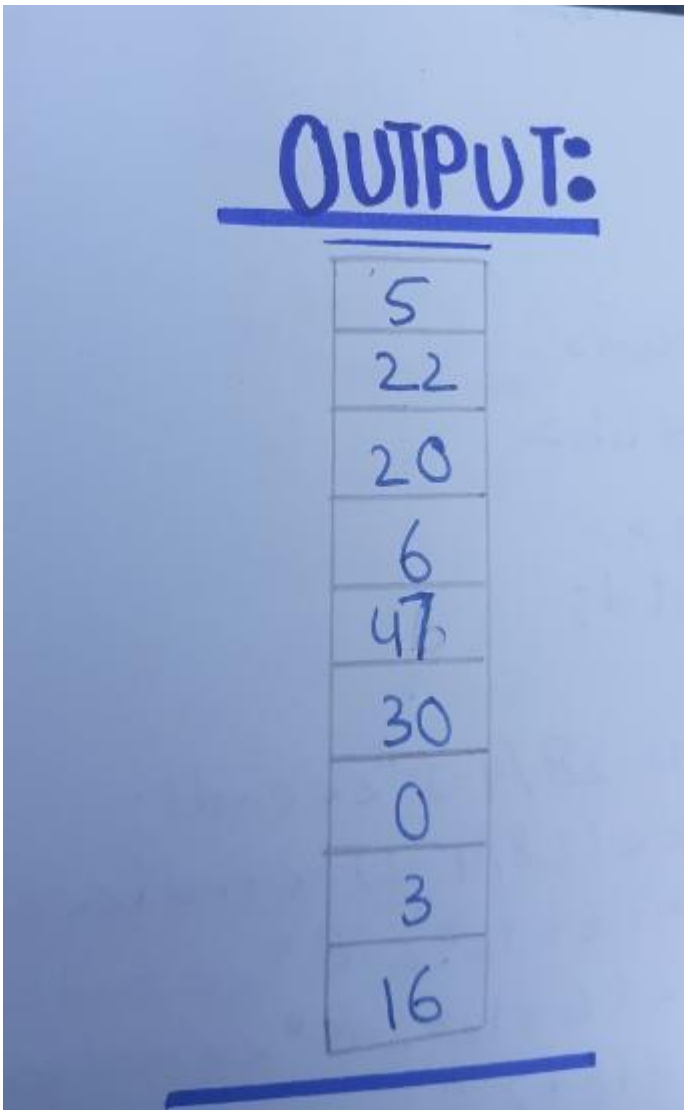
TASK. 2

Complete the following by writing the value of each expression in value coloumn according to C++ language.

Expression.	Value
28/4-2.	
6+12*2-8	
4+8*2	
6+17%3-2	
2+22*(9-7)	
(8+7)*2	
(16+7)%2-1	
12/(10-6)	
(19-3)*(2+2)/4	

HAND. WRITTEN CODING





CODING. ON PHONE

new*

```
#include <iostream>
#include <cstdlib>
#include <ctime>
using namespace std;
int main (){
cout<<"Expression:28/4-2"<<endl;
cout<<"value:"<<(28/4-
2)<<endl<<endl;
cout<<"Expression:6+12*2-8"<<endl;
cout<<"value:"<<(6+12*2-
8)<<endl<<endl;
cout<<"Expression:4+8*2"<<endl;
cout<<"value:"<<(4+8*2)<<endl<<endl;
cout<<"Expression:6+17%3-2"<<endl;
cout<<"value:"<<(6+17%3-
2)<<endl<<endl;
cout<<"Expression:3+22*(9-7)"<<endl;
cout<<"value:"<<(3+22*(9-
7))<<endl<<endl;
cout<<"Expression:(8+7)*2"<<endl;
cout<<"value:"<<((8+7)*
2)<<endl<<endl;
cout<<"Expression:(16+7)%2-
1"<<endl;
cout<<"value:"<<((16+7)%2-
1)<<endl<<endl;
cout<<"Expression:12/(10-6)"<<endl;
cout<<"value:"<<(12/(10-
6))<<endl<<endl;
cout<<"Expression:(19-3)*(2+2)/
4"<<endl;
cout<<"value:"<<((19-3)*(2+2)/
4)<<endl<<endl;

  return 0;
}
```

OUTPUT

TAB

```
Expression:28/4-2
value:5

Expression:6+12*2-8
value:22

Expression:4+8*2
value:20

Expression:6+17%3-2
value:6

Expression:3+22*(9-7)
value:47

Expression:(8+7)*2
value:30

Expression:(16+7)%2-1
value:0

Expression:12/(10-6)
value:3

Expression:(19-3)*(2+2)/4
value:16

[Program finished]
```

TASK. 3

Assume a program has following variable defined

```
int  unit ;
float mass;
double  weight;
weight= unit* mass;
```

Which automatic data type conversion will take place?

(A) mass is donated to an int,unit remain an int,and result mass*unit is int.

(B) mass is donated to an float,int donated to float
and reusult mass*unit is float.

(C) unit is donated to float, mass remain float and
then result mass*unit is double.

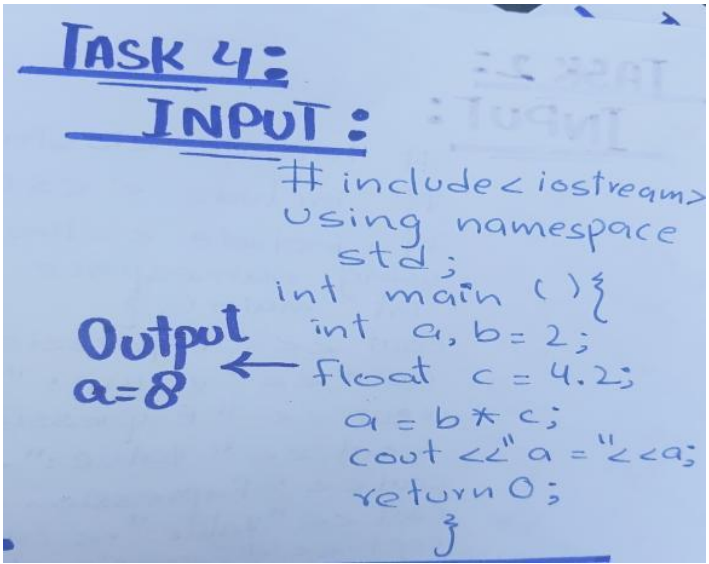
OUTPUT

B

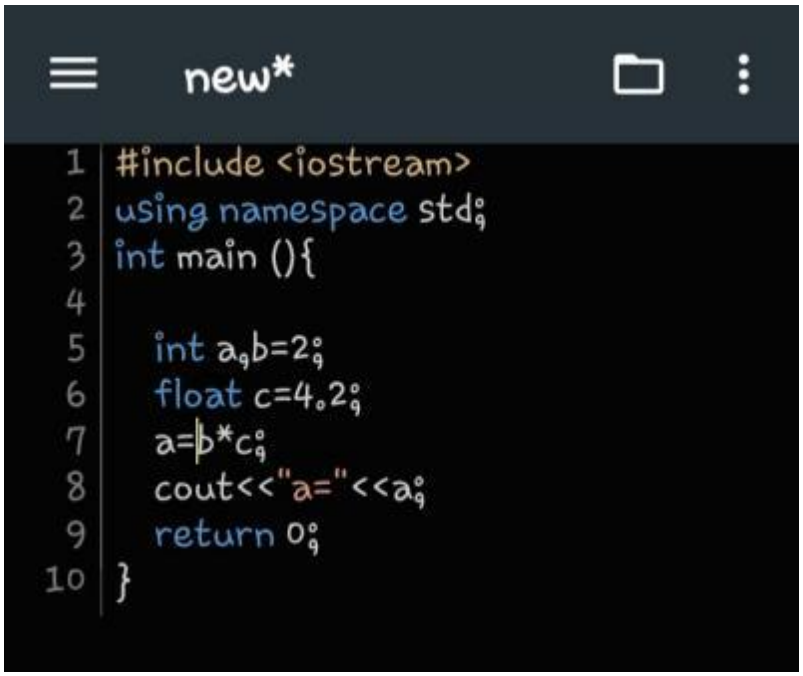
TASK. 4

Assume a program has following variable definition.
int a,b =2; float c=4.2;
and following statement is a=b*c:
A.8.4. B.8. C.0. D.None of these.

HAND WRITTING CODING.



CODE WRITTEN ON PHONE



OUTPUT



TASK 6

Math tutor (hint:rand())

write a program that can be used as math tutor for young student . The program should display two random number to be added then.

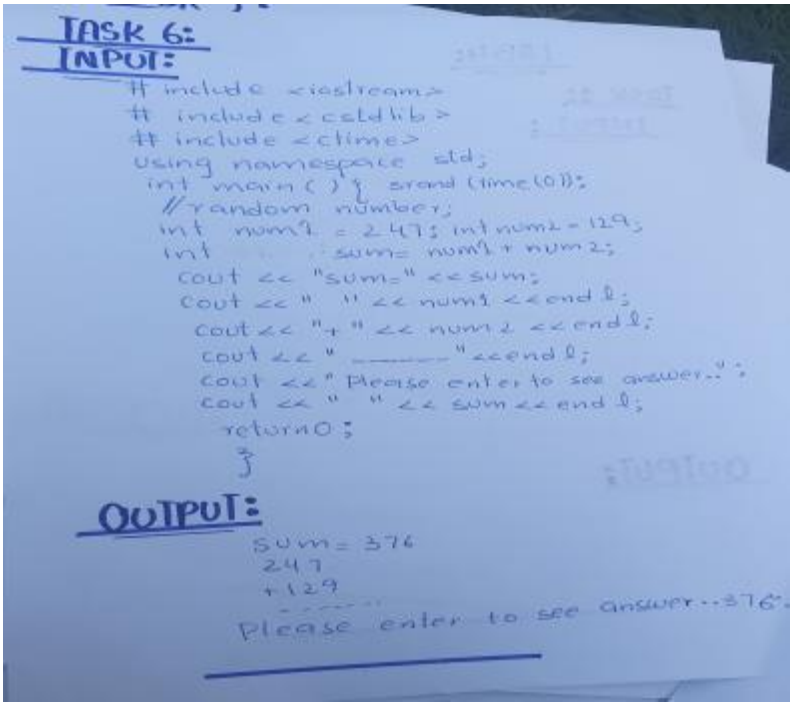
```
    247
 +  129
-------------
```

Ghe program should then pause while student work on problem.When student is ready to check the answer, he or she can press a key then solution will display on screen

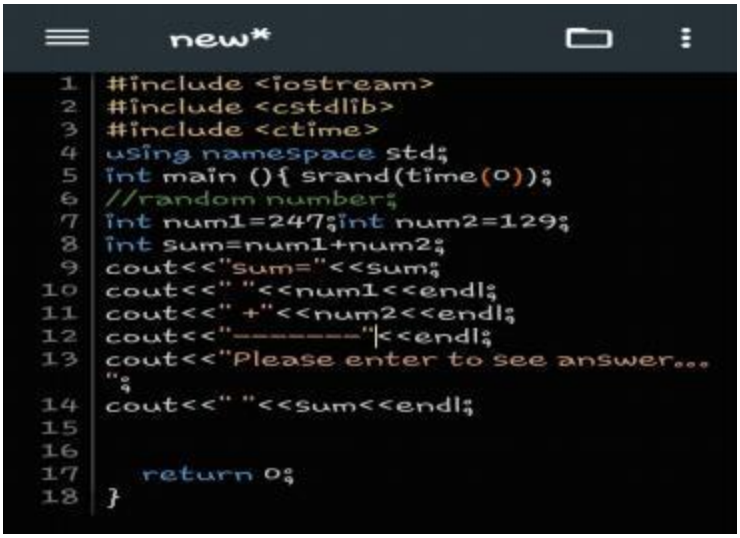
247
+ 129

376

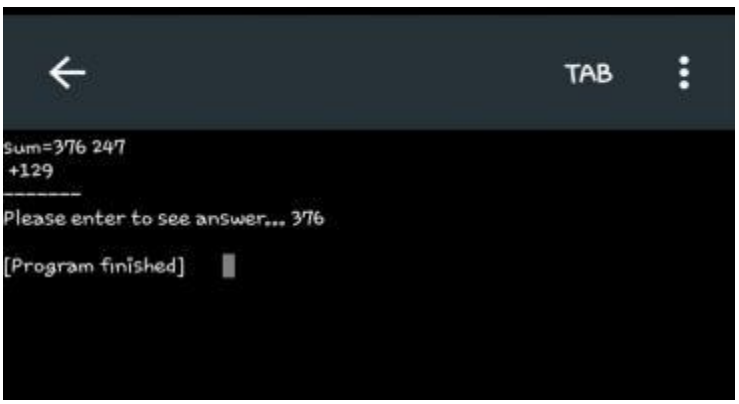
HAND WRITTEN CODING



CODING ON PHONE



OUTPUT

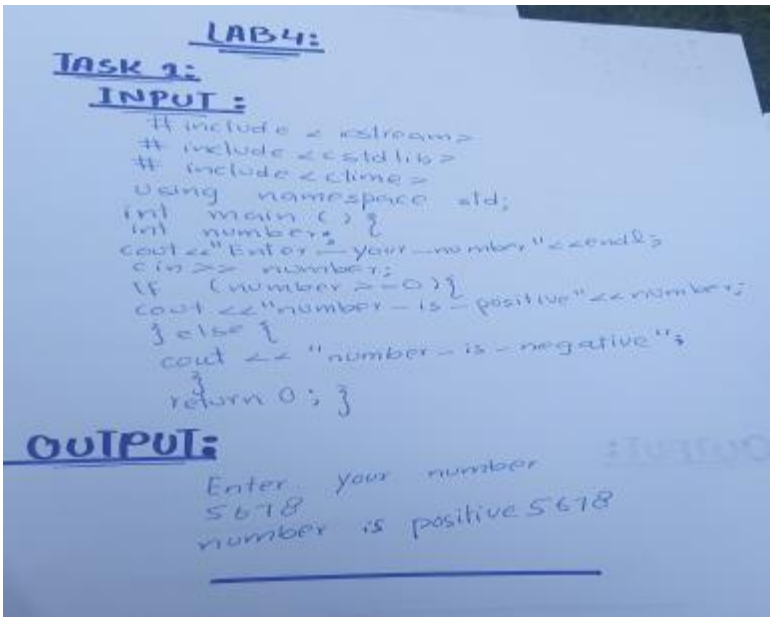


LAB. 4

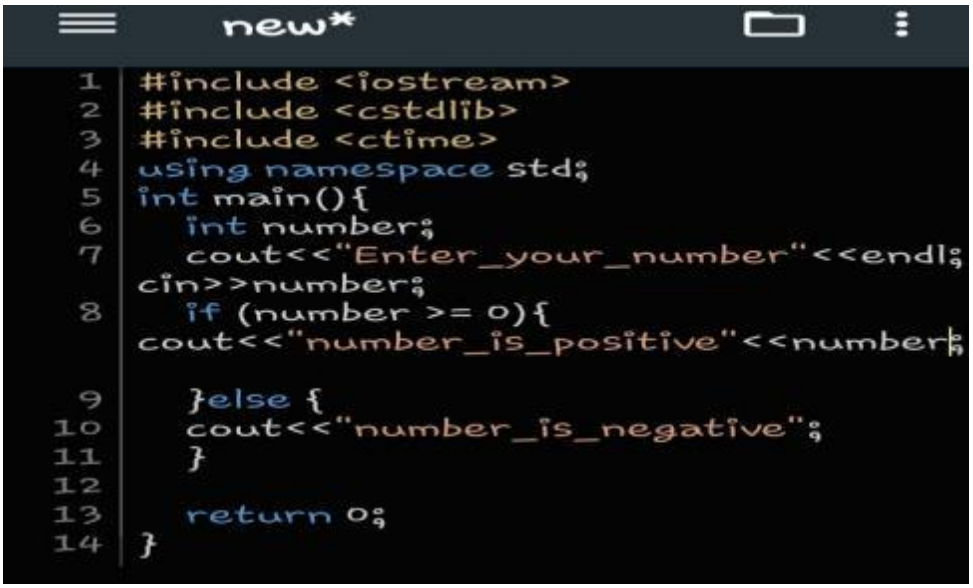
TASK. 1

Write program to print positive entered by user
if user enter negative number print number entered is positive otherwise print is negative.

HAND WRITTEN CODING



CODING ON PHONE



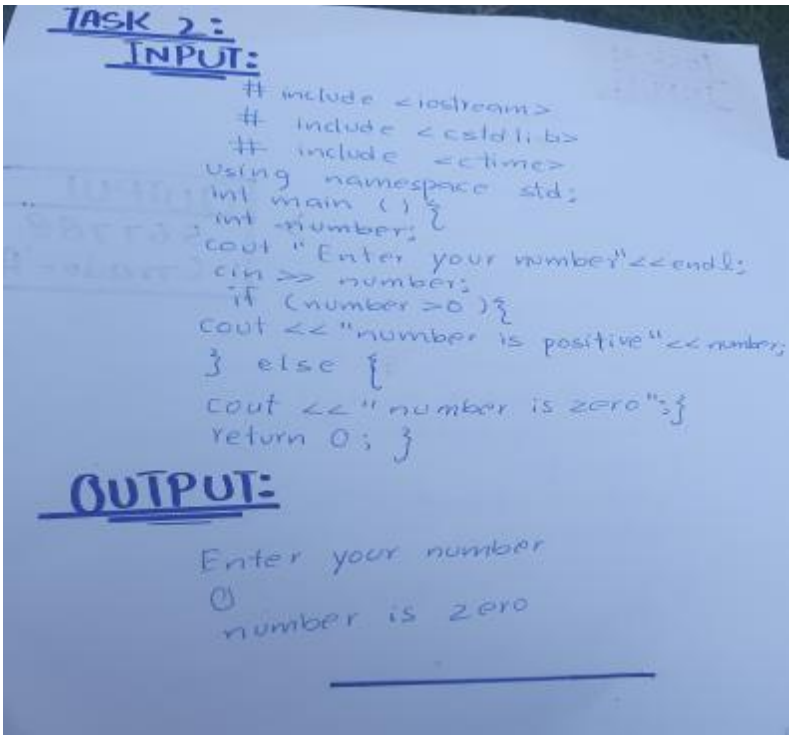
OUTPUT



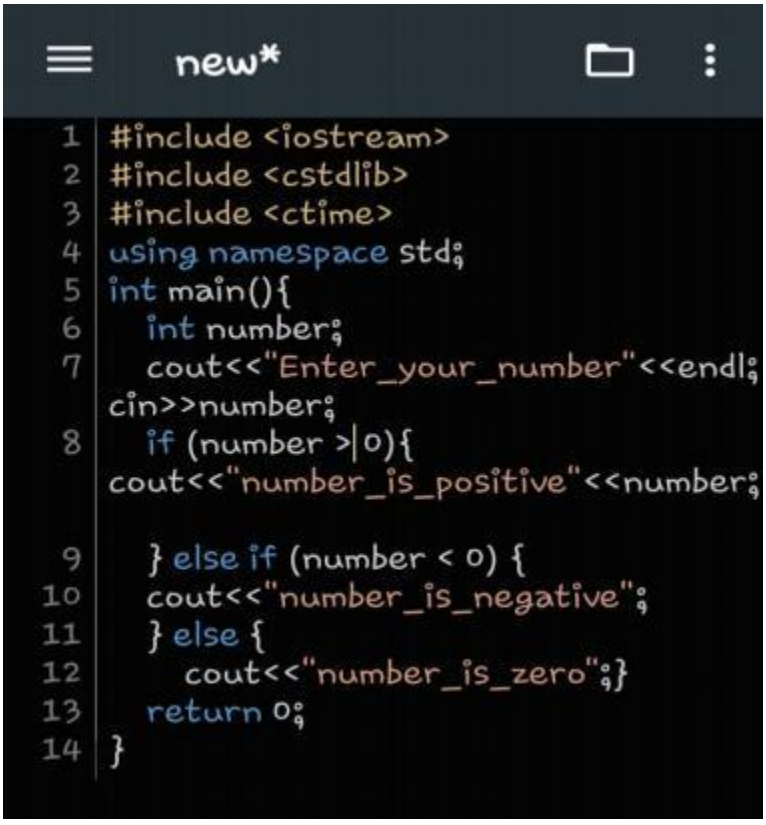
TASK. 2

Progrm check whether int is positive or negative or zero.

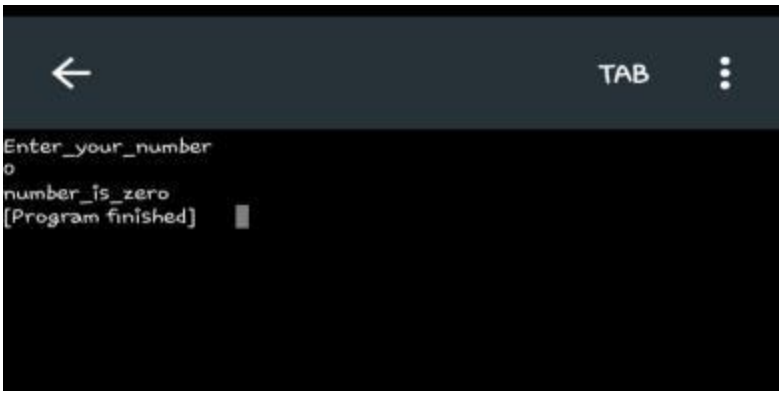
HAND WRITTEN CODING



CONDING WRITTEN ON PHONE



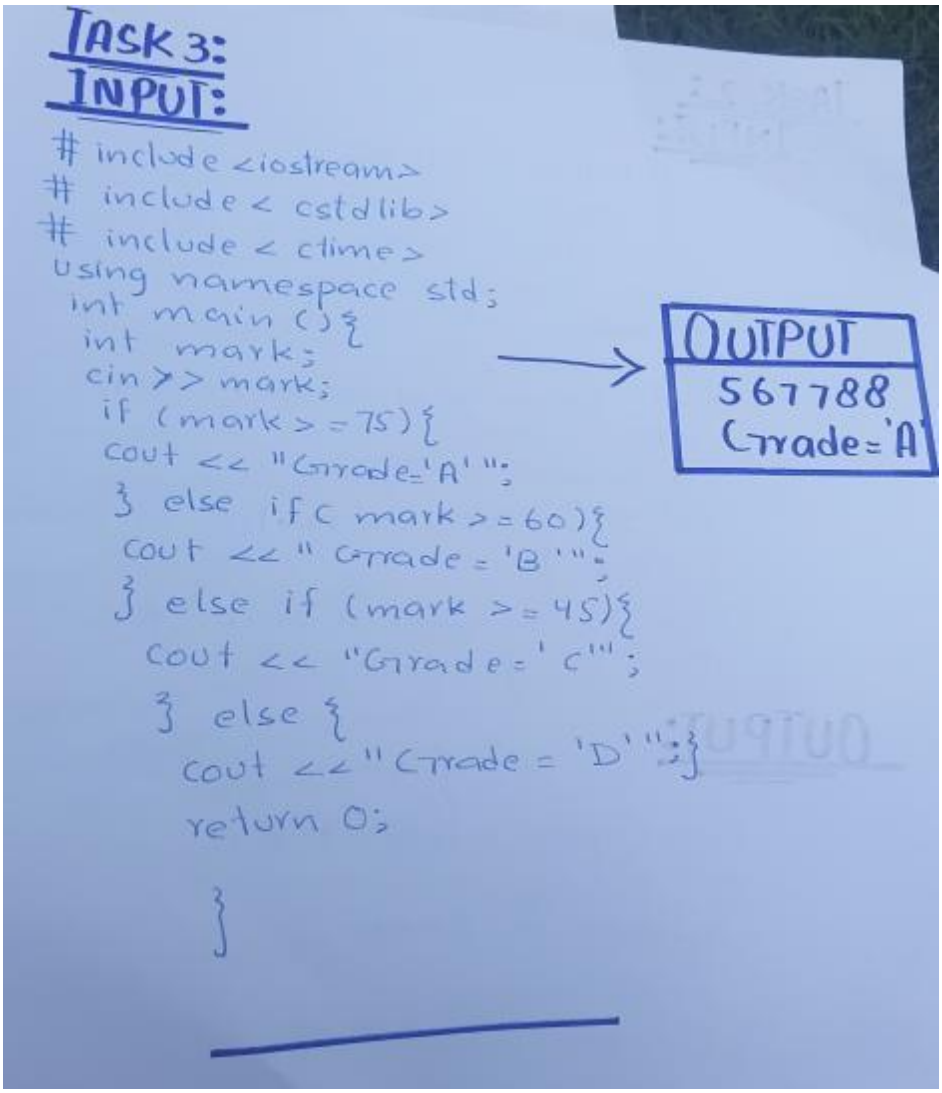
OUTPUT



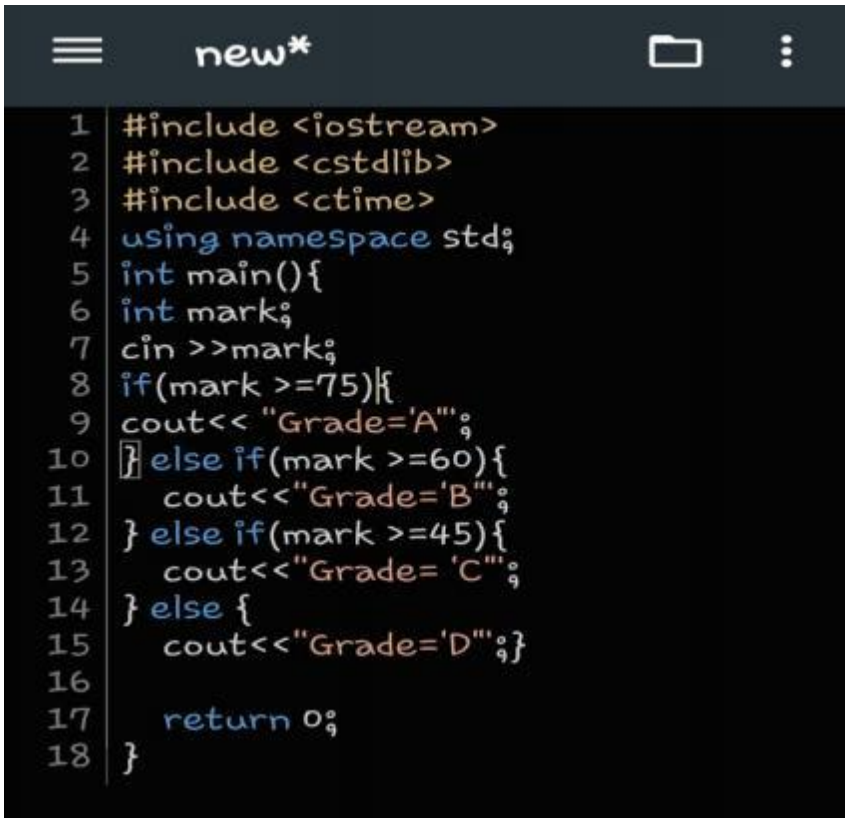
TASK 3

Int = mark
process:if mark greater than and equal to 75 , score will be A.
if mark less than 75 and greater than and equal to 60 ,score wiĺ be B.
if mark less than 60 and greater than and equal to 45 ,score will be C.
if mark less than 30 , score will be D.
Output :print grade of your score

HAND WRITTEN CODING



CODING WRITTEN ON PHONE



```
new*
1  #include <iostream>
2  #include <cstdlib>
3  #include <ctime>
4  using namespace std;
5  int main(){
6  int mark;
7  cin >>mark;
8  if(mark >=75){
9  cout<< "Grade='A'";
10 } else if(mark >=60){
11   cout<<"Grade='B'";
12 } else if(mark >=45){
13   cout<<"Grade= 'C'";
14 } else {
15   cout<<"Grade='D'";}
16
17   return 0;
18 }
```

OUTPUT



```
TAB
567788
Grade='A'
[Program finished]
```

TASK. 4

First largest number using nested ___ if else statement
Check whether number is affected by user is positive or not .
If it is positive then calculate how many digit number have

HAND WRITTEN CODING

TASK 4:

INPUT:

```
#include <iostream>
using namespace std;
int main (){
int number;
cout << " Enter number:";
cin >> number;
if (number>0) {
cout << "number is positive";
}
if (number<10){
cout << "digit1" <<endl;
}
  if (number < 100){
  cout <<"digit 2" <<endl;
}else if (number<1000){
 cout<< "digit 3"<<endl;
  } else {
    cout << "More than 3 digits";
  }
 cout << "nomber is not positive";
   return 0; }
```

OUTPUT:

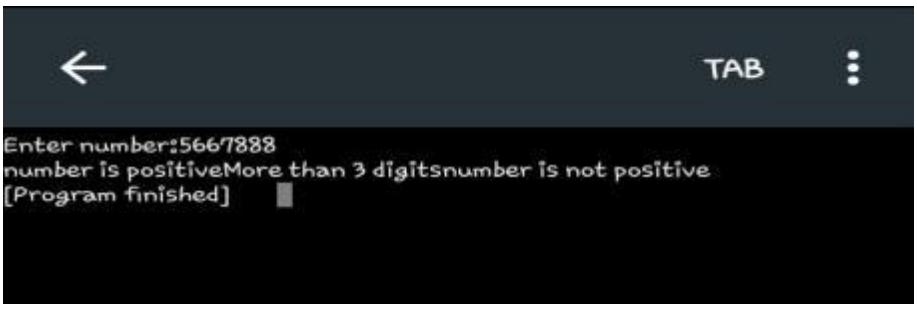
Enter number : 5667888

number is positive more than 3 is not positive.

CODING WRITTEN ON PHONE

```
1  #include <iostream>
2  using namespace std;
3  int main(){
4  int number;
5  cout<<"Enter number:";
6  cin>>number;
7  if(number>0){
8  cout<<"number is positive";
9  }
10 if(number < 10){
11    cout<<"digit1"<<endl;
12 }
13    if(number < 100){
14    cout<<"digit2"<<endl;
15 } else if(number < 1000){
16    cout<<"digit3"<<endl;
17 } else {
18    cout<<"More than 3 digits";
19 }
20    cout<<"number is not positive";
21
22    return 0;
23 }
```

OUTPUT



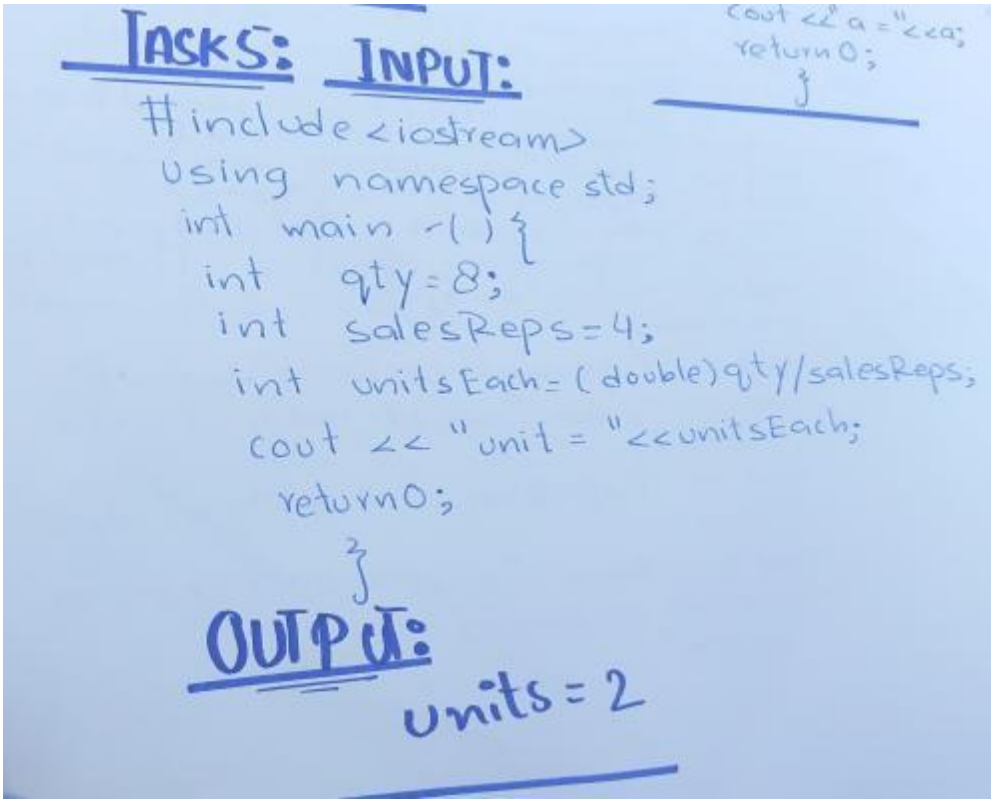
LAB. 3

TASK 5

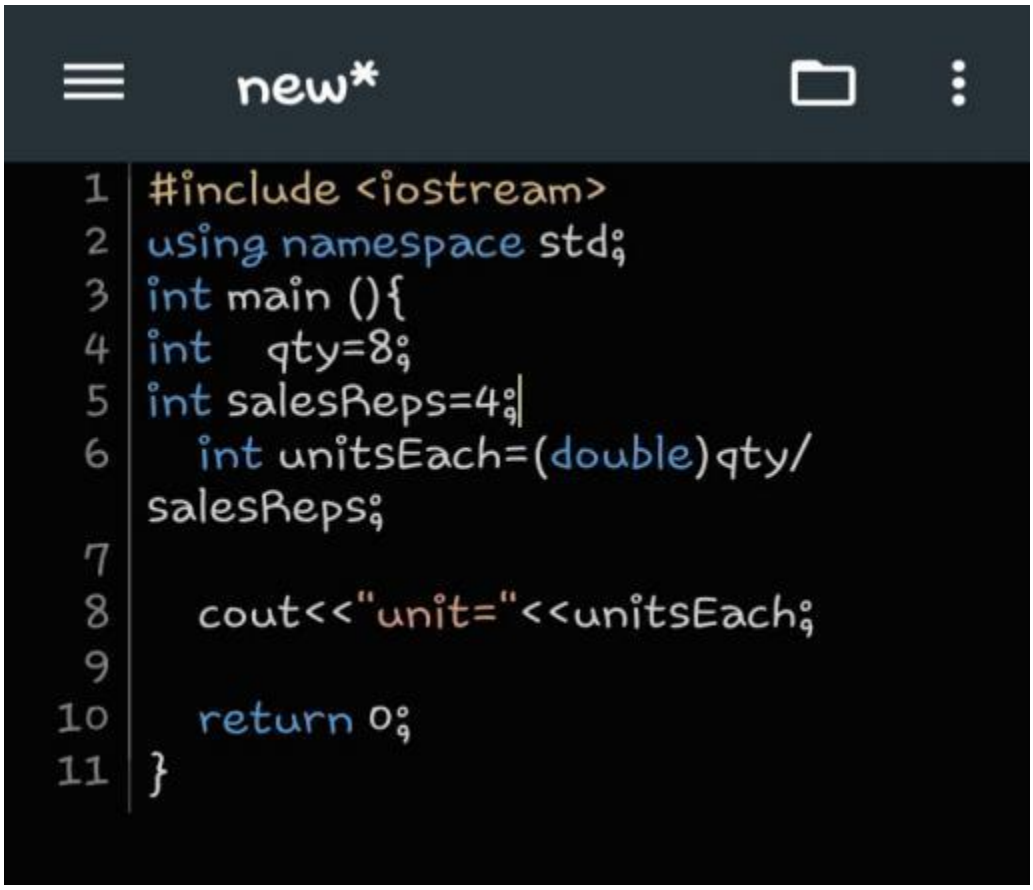
Assume qty and salesReps are both integer . Use a pdf can expression to rewritten following statement so it will not perform integer division

unitEach=qty/salesReps.

HAND WRITTEN CODING



CODING WRITTEN ON PHONE



OUTPUT

